

Data Oriented Systems

Home Assignment 2

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E-Procurement Data Pipeline

HW 2.2 - European Public Procurement
Multi-Source Data Integration

Problem

Challenge Context

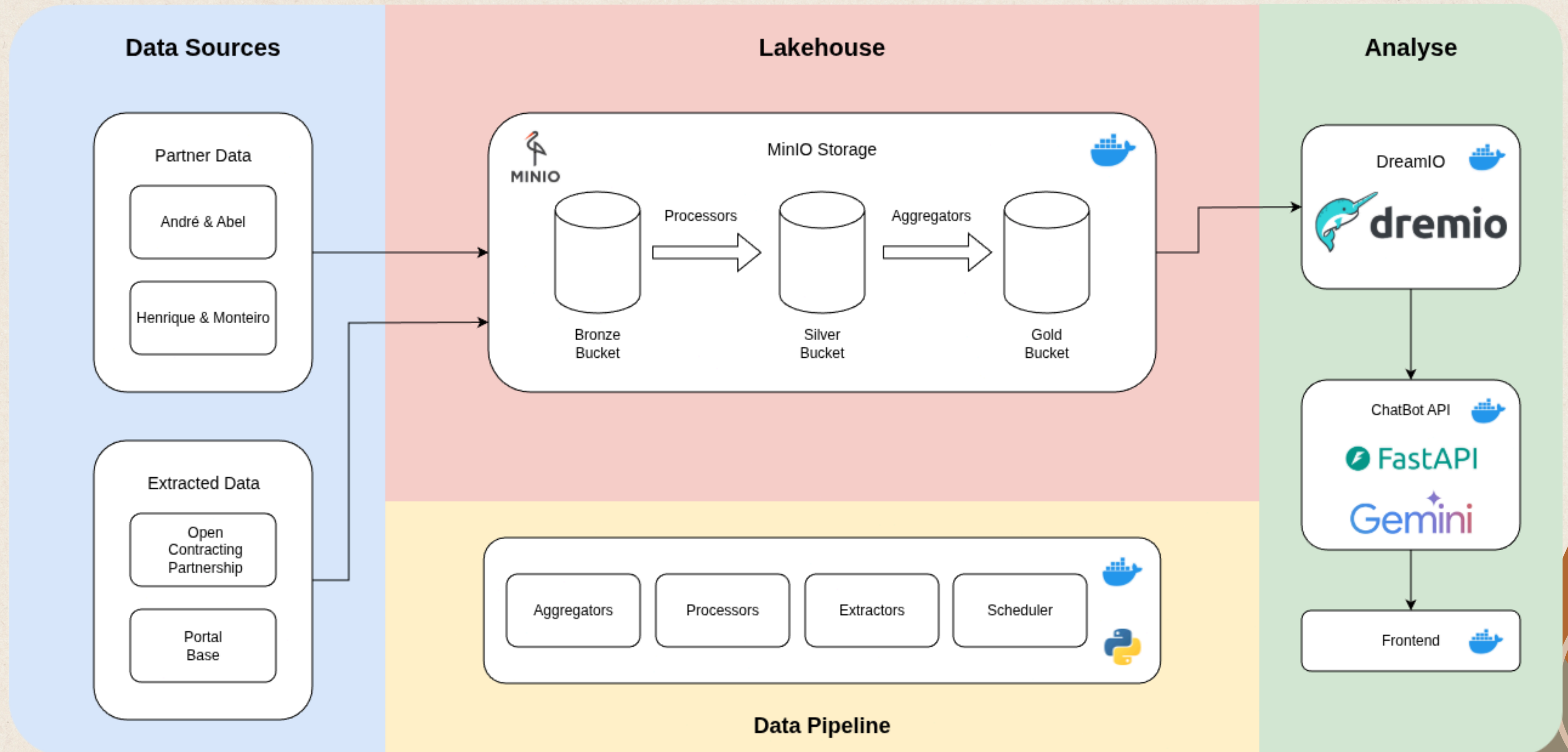
- European public procurement data is fragmented across multiple sources
- Different data formats, schemas, and quality levels create integration barriers
- Accessing and analyzing procurement data requires technical expertise
- Lack of accessible tools for transparency analysis and market insights

Key Pain Points

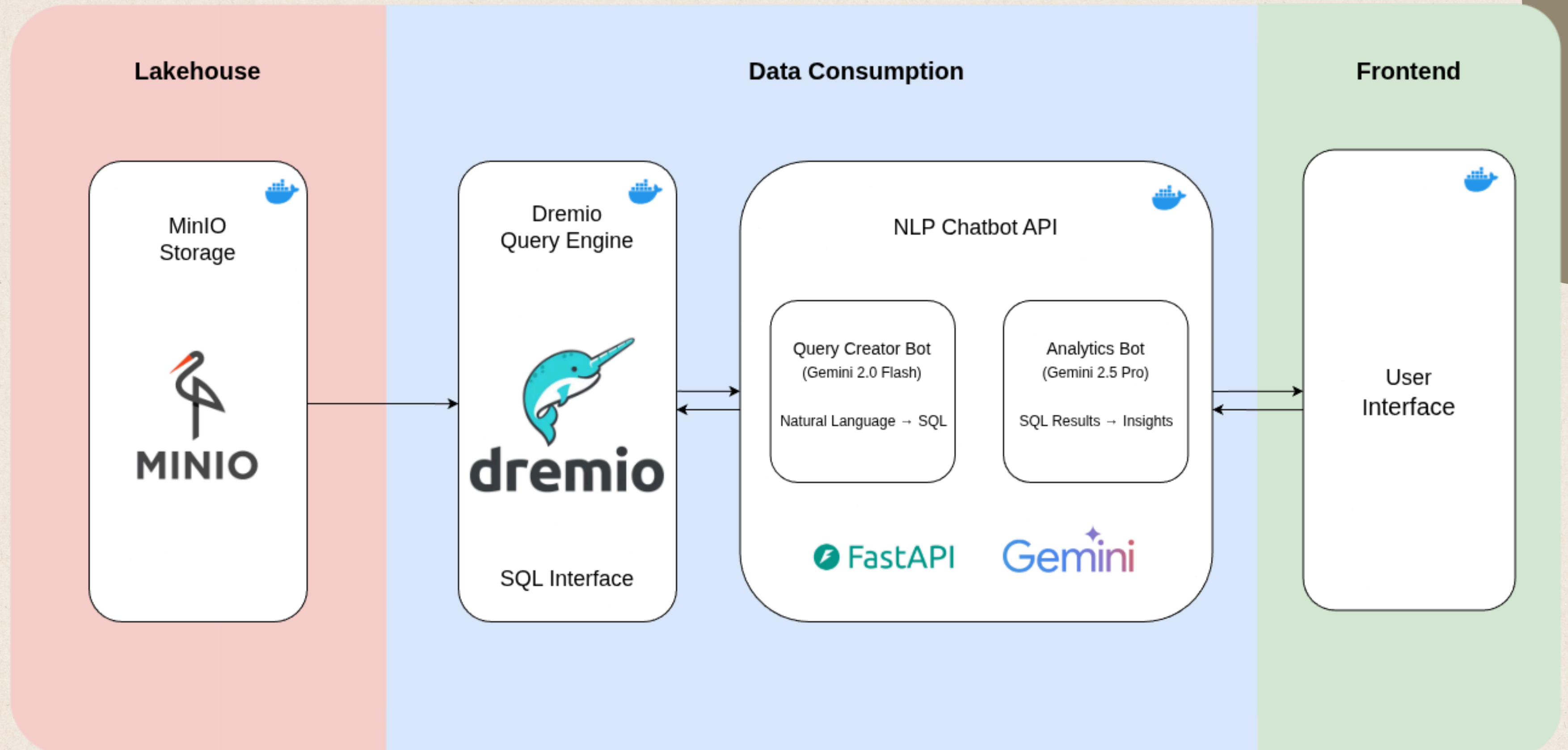
- Data Silos: Procurement information scattered across disparate systems
- Technical Barriers: Non-technical users cannot easily query the data
- Integration Complexity: Heterogeneous data sources need standardization
- Limited Accessibility: No unified interface for procurement analytics



System Architecture



ChatBot



Live Demo

Conclusion

What We Achieved:

- Built a complete end-to-end data pipeline integrating multiple European procurement sources
- Implemented a medallion architecture (Bronze → Silver → Gold) for progressive data refinement
- Created an NLP-powered chatbot enabling natural language queries for non-technical users
- Democratized access to public procurement data through an intuitive interface

Future Work:

- Expand to additional European countries and data sources
- Enhance ML models for procurement anomaly detection
- Add visualization dashboards for trend analysis

